

Implementing an NTP Client in Ubuntu Linux



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Overview



Network Time Protocol

- timedatectl
 - timezone
 - set-ntp
 - timesync-status
 - show-timesync
 - systemd-timesyncd
- Configuring NTP client
 - `/etc/systemd/timesyncd.conf`



NTP: Network Time Protocol,
used to keep accurate time on
systems



```
$ timedatectl  
$ sudo timedatectl set-timezone 'Europe/London'  
$ systemctl status systemd-timesyncd  
$ sudo timedatectl set-ntp true  
$ timedatectl  
$ sudo systemctl status systemd-timesyncd
```

Systemd-Timesyncd

By default, Ubuntu makes use of the NTP client-only systemd-timesyncd; giving systems a smaller footprint and increased security. The NTP client can be managed via timedatectl.

Demo



Configuring Time:

- timedatectl
- timezone
- ntp client



Whilst the default time server is adequate, servers often will synchronize with an infrastructure device within the data-centre



```
$ timedatectl timesync-status
$ timedatectl show-timesync
$ sudo vim /etc/systemd/timesyncd.conf
$ sudo timedatectl set-ntp false && sudo timedatectl set-ntp true
$ timedatectl timesync-status
```

Setting NTP Peers

Configuring the NTP server to sync with, is a simple edit of the configuration file. Again, we can manage this without the need of using systemctl to restart the service.

Demo



Configuring NTP Peers:

- `/etc/systemd/timesyncd.conf`
- `timesync-status`
- `--monitor`
- `show-timesync`



Summary



Time and NTP

- timedatectl
- systemd-timesyncd
- /etc/systemd/timesyncd.conf
- set-ntp
- set-timezone
- timesync-status



Configuring Network Routing in Linux

